Pearly Lin Lee Ying

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Portfolio: **https://pearlylin2004.github.io/Portfolio/**



PROFESSIONAL SUMMARY

Computer Science student with a strong foundation in full-stack development and robotic process automation(RPA) and Data Engineering. Proven track record of delivering high-impact automation solutions, including saving 300 man-hours annually through dynamic query systems. Experienced in Python, Java and C++ with a focus on optimizing runtime performance, building custom engines and implementing complex state management systems.

FEATURED EXPERIENCE

Software Engineer Intern | National Youth Council (NYC) | *Sep 2023 – Aug 2024*

Supervisor: Hazlami Zawawi (then Manager, now Republic Polytechnic IT Dept) • View Testimonial

- **Software Automation:** Developed and optimized a dynamic query system for FormSG responses using M language, automating data storage and saving 300 man-hours per year.
- **Performance Engineering:** Refactored data transformation queries for email marketing, significantly reducing runtime and enabling the processing of thousands of records monthly.
- **RPA Development:** Enhanced financial and grant databases bot using UIPath, implementing advanced exception handling to ensure operational stability.
- **Project Leadership:** Led a cross-institutional project with Nanyang Polytechnic to automate account creation, accelerating the commissioning timeline through technical leadership.
- **Quality Assurance:** Collaborated on a comprehensive code review checklist to establish quality standards and best practices for current and future automation software.

PROJECTS

WaterBound(TanukiLab) – Custom Engine & Framework | Programmer & Collision Champion | *Sept 2025 – Feb 2026*

- **Core Systems & Architecture:** Developed a custom software engine in C++ featuring component-based architecture and designed a robust, optimized Collision Detection System.
- **OS & Event Integration:** Engineered low-level Windows lifecycle and interruption handling (e,g,. implementing window focus loss detection and automated pause states on system interrupts like Ctrl-Alt-Del)
- **Technical Debt & QA:** Voluntarily stepped up to refactor, debug and sanitize team-contributed code across physics and core state management systems, ensuring final application stability for a 9-person team
- **Technical Communication:** Collaborated with the team to deliver technical project updates, effectively communicating architecture choices and engine capabilities to stakeholder.

CandyDefender6000 – Strategy System | Systems & Level Programmer | Team of 4 *Jan 2025 – Mar 2025*

- **State Management:** Implemented core game state transitions and level progression logic in C++ to handle entity updates and win/loss conditions
- **System Troubleshooting:** Performed extensive debugging and diagnostic testing across multiple gameplay levels to isolate and resolve software defects.
- **Level Integration:** Configured level-specific parameters and integrated dynamic entity states to ensure stable runtime execution.

DunkinDonut (Object-Oriented Project) | Software Developer | Team of 6 *Sept 2024 – Dec 2024*

Object-Oriented Design: Implemented core logic for player growth and collision scaling in C++ using Object-Oriented Programming principles.

- **Entity Management:** Developed centralized entity management systems and pattern-based state behaviours.

WORK EXPERIENCE

Kitchen Crew (Part-Time) | Hong Le Ptd Ltd *Mar 2021 – Dec 2025*

- **Operational Efficiency:** Independently managed complex kitchen stations and executed workloads typically allocated to a 3-person team, maintaining high-quality outputs under pressure.
- **Crisis Management and Reliability:** Trusted by management as a critical team resource, regularly called upon on short notice to resolve staffing shortages and maintain seamless operations.
- **Cross-Cultural Communication:** Leveraged multilingual skills to serve as a key workplace interpreter for international team members, bridging communication gaps and ensuring operational safety and clarity.
- **Time Management:** Multitasked across high-volume orders simultaneously to guarantee consistent delivery within a strict 10-minute KPI

LEADERSHIP & ACTIVITIES

Team Member | Dragon Boat, Singapore Institute of Technology *Sept 2024 – Current*

- Developed high physical endurance, discipline and synchronized team execution through rigorous training regimes and competitive racing

Volunteer | Citizen on Patrol (Singapore Police Force) *Sept 2024 – Current*

- Partnered with local law enforcement to conduct regular neighbourhood patrols, contributing to community safety and crime prevention initiatives.

Volunteer | Resident's Network Observer (People's Association) *Oct 2025 – Current*

- Participate in monthly strategic meetings to contribute to community roadmap planning: activity organize, plan and execute neighbourhood engagement activities.

Ambassador | Silver Generation Office *2024 - Current*

- Conduct regular outreach and check-ins with senior citizens, leveraging active listening and multilingual communication skills to understand their needs and share relevant community support schemes

Peer Supporter | Republic Polytechnic *2022 – 2024*

- Appointed during year 2 to anchor a class of 20 junior students continuously from their Year 1 induction to Year 3 graduation; voluntarily extended mentorship post-graduation to provide ongoing guidance, active listening and academic tracking.

Vice–President | Tabletop Group, Republic Polytechnic *2023 – 2024*

- Oversaw member safety, event participation and strategic planning weekly sessions

Vice–President (Staff Sergeant) | Girls Brigade: Received the Best Cadet Award(2020) *2017 – 2020*

- Represented the organization in national parades and managed large scale donation drive

EDUCATION

BSc (Hons) in Computer Science in Interactive Media and Game Development *Expected Graduation October 2028*
Singapore Institute of Technology, DigiPen Institute of Technology, Singapore
Relevant Courses: Data Structures & Algorithms, Computer Graphics, Game Implementation Techniques, Artificial Intelligence, User Interface Design, Software Engineering, Scripting Languages

Diploma in Information Technology *April 2021 – May 2024*
Republic Polytechnic
Awards: MOE Edusave certificate of Academic Achievement (Top 25% cohort)
Relevant Courses: Database Systems, Internetworking, Operating Systems and Virtualization

TECHNICAL SKILLS

Languages: C, C++, JavaScript, Java, Python, C#, M Language, SQL
Tools & Frameworks: UIPath (RPA), Unity, Dear IMGui, Git, GitHub, Power Query, Database System Virtualization
Concepts: System Optimization, State Management, Object-Oriented Programming (OOP), Data Transformation, Code Review, Exception Handling, Windows API Interruption Handling
Areas of Interest: Civic Technology, Public Sector Automation, System Integration, Application Development, Cyber-Physical Systems